香港中文大學（深圳）
The Chinese University of Hong Kong, Shenzhen

DDA2001: Introduction to Data Science

Lecture 7: Continuous Random Variable

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Terminologies

- **Random Experiment**: a repeatable procedure
- **Sample Space**: set of all possible outcomes S.
- **Probability function**, $P(\omega)$: how likely the outcome $\omega \in S$
    - Probability is between 0 and 1
    - Total probability of all possible outcomes is 1.

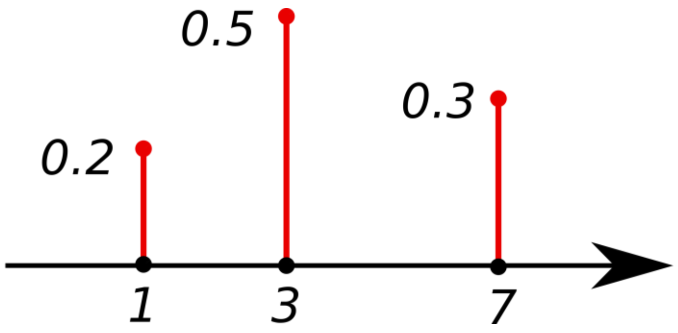
Sample space

- Discrete or continuous: countable (listable) or not?

- A sample space is **discrete** if it consists of a finite or countable infinite set of outcomes.
- A sample space is **continuous** if it contains an interval (or a union of multiple intervals) of real numbers.

Probability function

- Discrete:
  - ✓ Probability mass function.
  - ✓ P(s): gives the probability for each outcome $\omega \in S$



Recap of Common Discrete RV

1.Bernoulli distribution

- Take value 1 with probability p and value 0 with probability $1 - p$.

- $X \sim Bernoulli(p)$

- Mean=p
- Variance=p(1-p)

2.Binomial distribution

- N independent experiments
- Each experiment: success (with probability p) or failure (with probability $1 - p$).
- X: the number of success (failure).
- $X \sim Binomial(N, p)$
- Mean=Np
- Variance=Np(1-p)

3.Geometric distribution

- Continuously draw a Bernoulli R.V.
- The X-th sample is the first success.
- X follows a geometric distribution.

- $X \sim Geometric(p)$ or $X \sim Geo(p)$

- Mean=1/p
- Variance=(1-p)/p^2

Continuous Random Variable

Suggested reading material

- Applied Statistics and Probability for Engineers, Third Edition, Douglas C. Montgomery and George C. Runger.
  - Chapter 4.4-4.6, 4.8

- Havard Stat 110, book by Joe Blitzstein and Jessica Hwang
  - https://drive.google.com/file/d/1VmkAAGOYCTORq1wxSQqy255qLJjTNvBI/edit Chapter 5.1-5.4

Motivation Example

True or not?

- If $P(E) = 0$, then it is impossible to observe E.

True or not?

- If $P(E) = 0$, then it is impossible to observe E.

False

What's the meaning of impossibility?

- **Random Experiment**: a repeatable procedure
- **Sample space**: set of all possible outcomes S.
- An event is a subset of possible outcomes.

- If P(E) = 0, then E is a zero-probability event.
- If E is empty, then E is impossible.

Is a zero-probability event impossible?

Example

- We have a probability model.
  - X represents the time a machine needs to complete a task.
  - The values X can take are within [0,1].
  - For any $x, y \in [0,1]$, we have $P(X = x)$= $P(X = y)$
  - E = {0.5}
  - Then what's the probability of $P(E)$?

Example

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Although the impossible event has zero probability, not all zero-probability events are impossible.

Example

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  - For any $x, y \in [0,1]$, we have $P(X = x)$= $P(X = y)$
  - E = [0, 0.5]
  - Then what's the probability of $P(E)$?

General Case

- X is a random variable.
- Define its range as S.
- For any $x, y \in S$, we have $\frac{P(X=x)}{P(X=y)} = \frac{f(x)}{f(y)}$.
  - $\boldsymbol{f(x) > 0}$ **for any** $x \in S$**.**
  - $\boldsymbol{f(x) = 0}$ **for any** $x \notin S$**.**
- E = [a, b] $\subseteq S$
- Then what's the probability of $P(E)$?

General Case

An intuitive interpretation, but not rigorous

- X is a random variable.
- Define its range as S.
- For any $x, y \in S$, we have $\frac{P(X=x)}{P(X=y)} = \frac{f(x)}{f(y)}$.
  - **$f(x) > 0$ for any $x \in S$.**
  - **$f(x) = 0$ for any $x \notin S$.**
- E = [a, b] $\subseteq S$
- Then what's the probability of $P(E)$?

$$\frac{\int_a^b f(x)dx}{\int_S f(x)dx}$$

Takeaways

- In continuous probability distributions, the probability of a **specific value** occurring is **always zero**. This is because there are infinitely many possible values the random variable can take, and each individual point has no "width" or "area" to assign a positive probability.
- **The density** tells you about the "likelihood" at that point, so the **ratio of densities** can tell you **which event is more likely relative to the other**, even though the probabilities themselves are zero.

Continuous RVs

1. How to describe?

Continuous R.V.

- A continuous random variable can take any value within its range (an interval of a union of multiple intervals of real numbers).
- We cannot list all the possible values and their probabilities as in the discrete case.

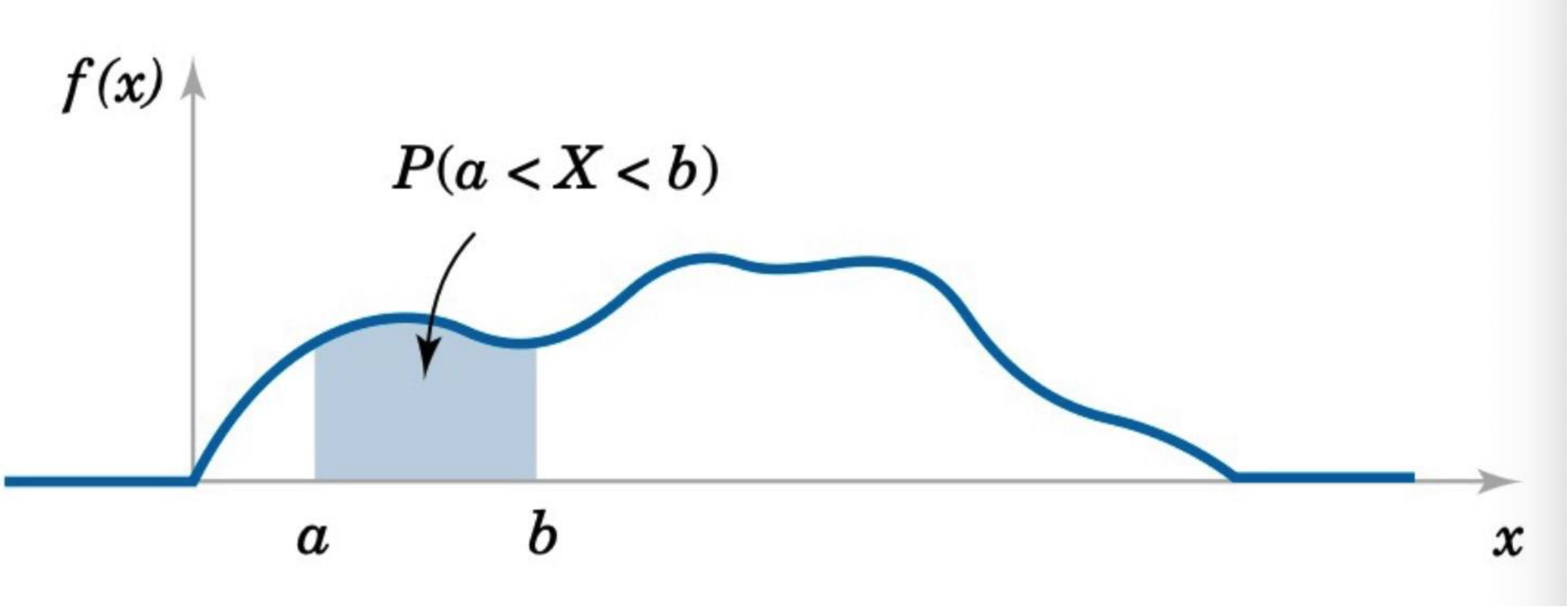
How to describe the probability?

- For a continuous RV X, P(X=x) = 0 but {x} is not an impossible event.
- We will not use the probability mass function (pmf), namely P(X=x).
- Instead, we introduce a function f(ω), called the probability **density** function (pdf).
  - f(ω) > 0, if $\omega \in S$
  - f(ω) = 0, if $\omega \notin S$
  - $\int_{-\infty}^{\infty} f(x)dx = 1.$

Probability of $X \in [a, b]$

In our course, for continuous RV, we mainly focus on the case where the range is only one interval, not a union of multiple intervals.

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$



Properties of PDF

- For *x* that is not in the sample space, *f(x)=0*

- A large value of *f(x)* means that the values around *x* is more likely to be observed. (remember this implication)

- As a pdf, f(x) can be larger than 1, while as a pmf, f(x) cannot be larger than 1.
  - f(ω) = 2 , if $\omega \in [0, 0.5]$
  - f(ω) = 0, if $\omega \notin [0, 0.5]$

Continuous RVs

2. CDF, Mean, and Variance

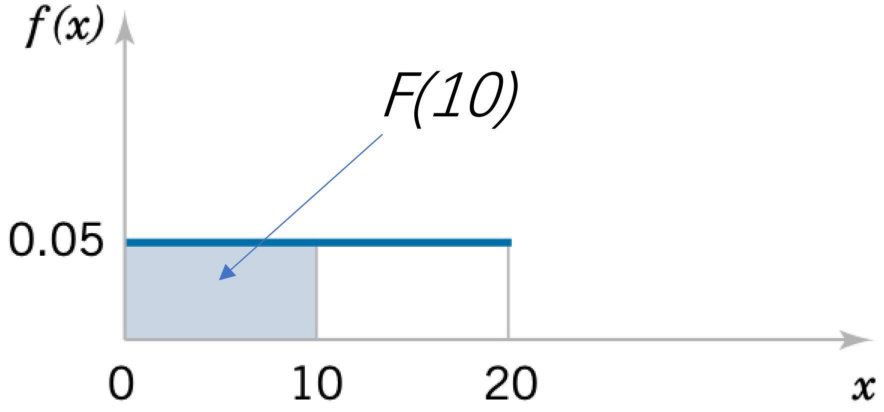
CDF

- Recall: the CDF of a discrete random variable X is

$$F(x) = P(X \leq x) = \sum_{\tilde{x} \leq x} f(\tilde{x})$$

- CDF for continuous random variable is defined as:

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(u)du$$



CDF

- Recall: the CDF of a discrete random variable X is

$$F(x) = P(X \leq x) = \sum_{\tilde{x} \leq x} f(\tilde{x})$$

- CDF for continuous random variable is defined as:

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(u)du$$

✓ $0 \leq F(x) \leq 1$

✓ If $x \leq y$, then $F(x) \leq F(y)$

For both discrete and continuous RVs

Mean and Variance

- Discrete:
  - ✓ Probability mass function.
- Continuous
  - ✓ Probability density function.

Summation ↔ Integration

- Mean

$$\mathrm{E}[X] = \sum x\, f(x)$$

- Variance

$$\mathrm{Var}[X] = \sum (x - \mathrm{E}[X])^2\, f(x)$$

- Mean

$$\mathrm{E}[X] = \int_{-\infty}^{\infty} x f(x) dx$$

- Variance

$$\mathrm{Var}[X] = \int_{-\infty}^{\infty} (x - E(X))^2 f(x) dx$$

Expectation of g(X)

$$\mathrm{E}[\mathrm{g}(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx$$

Continuous RVs

3. Common Distributions

Uniform Distribution

How to calculate $\int_0^2 e^{x^2+\cos(x)}dx$?

How to approximate $\int_0^2 e^{x^2+\cos(x)}dx$?

- We will demonstrate how to sample random variables from a **Uniform distribution** and use the sample mean to approximate the value of an integral.

- With the same probability, X takes a value within [a, b], where b>a.

Discrete version: toss a coin, roll a dice.

- What's the pdf?

- With the same probability, X takes a value within [a, b], where b>a.

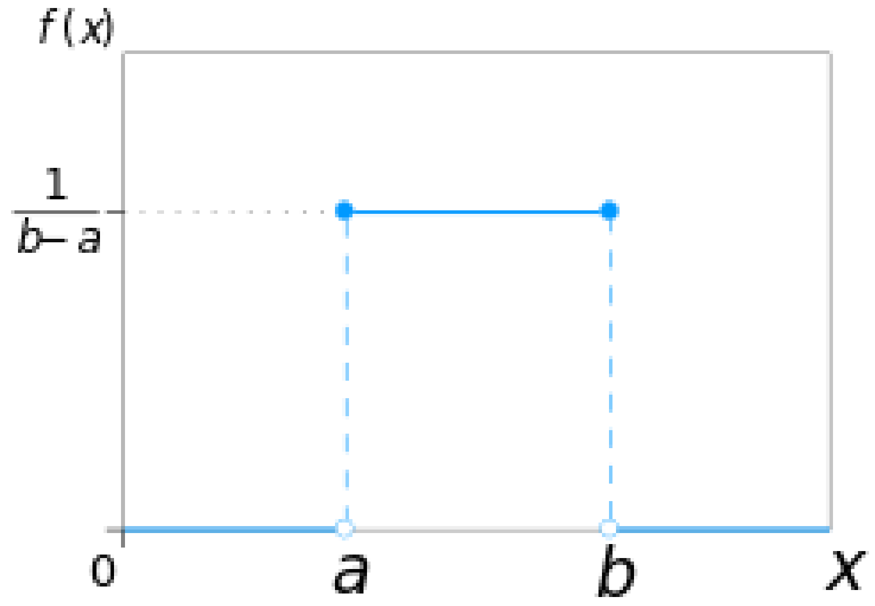
- What's the pdf?

- f(x) = c for $x \in [a, b]$ and f(x) = 0 for $x \notin [a, b]$
- As $\int_{-\infty}^{\infty} f(x)dx = c(b-a) = 1,$ we have

$$c = \frac{1}{b-a}$$

Uniform Distribution

- With the same probability, X takes a value within [a, b]
- $X \sim Uniform(a, b)$



Mean= $(a+b)/2$
Variance=$(b-a)^2/12$

Applications

- Given $X \sim Uniform(0,2)$
- What's the value of $E[2\, e^{X^2+\cos(X)}]$?

Applications

- Given $X \sim Uniform(0,2)$
- What's the value of $E[2\, e^{X^2+\cos(X)}]$?

- f(x) = ½ for $x \in [0,2]$
- $E[2\, e^{X^2+\cos(X)}] = \int_0^2 2\, e^{x^2+\cos(x)} f(x) dx = \int_0^2 e^{x^2+\cos(x)} dx$

How to approximate $\int_0^2 e^{x^2+\cos(x)}dx$?

Given $X \sim Uniform(0,2)$, $E[2\, e^{X^2+\cos(X)}] = \int_0^2 e^{x^2+\cos(x)} dx$

- Draw N samples of $X \sim Uniform(0,2)$: $X_1, X_2, X_3, \ldots, X_N$
- Calculate $\frac{\sum_i 2\, e^{X_i^2+\cos(X_i)}}{N}$

Why? Expectation can be approximated by long-run average.

General Case

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx$$

- Draw N samples of X: $X_1, X_2, X_3, \ldots, X_N$
- Calculate $\frac{\Sigma_i g(X_i)}{N}$

General Case

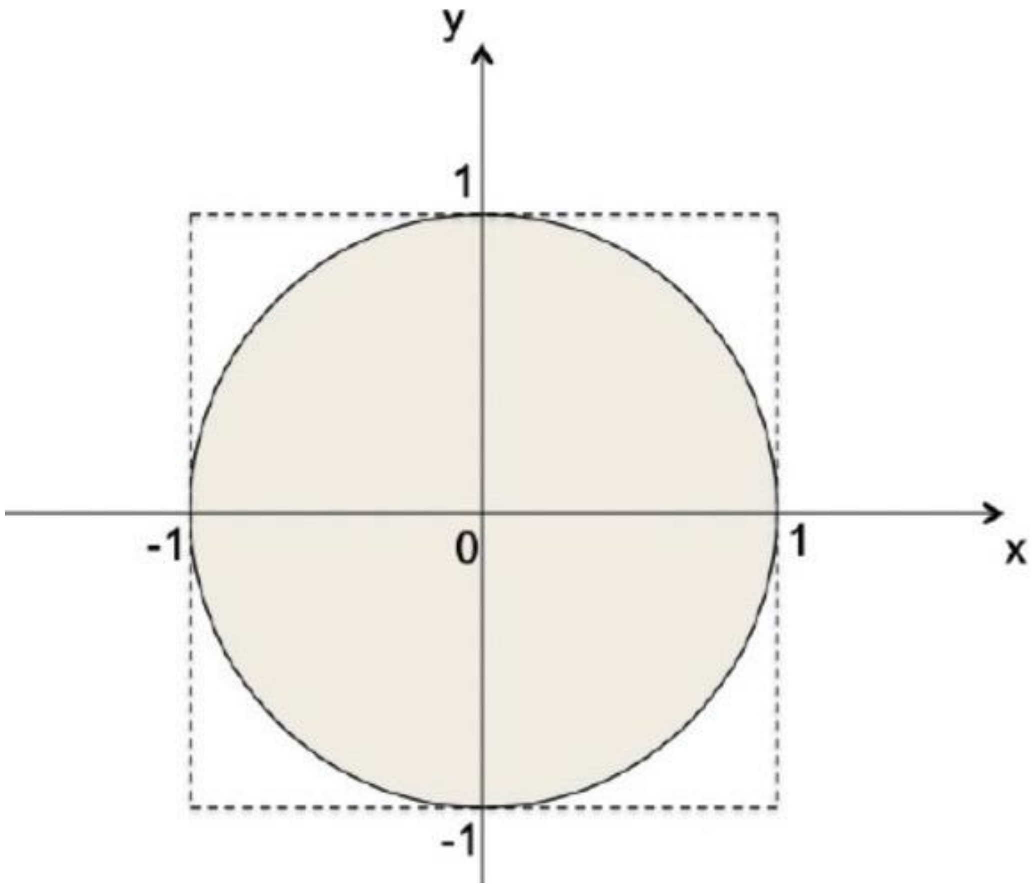
- How to calculate $\int_a^b h(x)dx$?

  - Draw N samples of X$\sim Uniform(a,b)$: $X_1, X_2, X_3, \ldots, X_N$
  - Calculate $\frac{\Sigma_i (\boldsymbol{b}-\boldsymbol{a})\, h(X_i)}{N}$

- Let $X \sim Uniform(a,b)$
- f(x) = 1/(b-a) for $x \in [a,b]$
- $E[(b-a)h(x)] = \int_a^b (b-a)h(x)f(x)dx = \int_a^b h(x)dx$

Exercise

- Estimation of π



Exercise

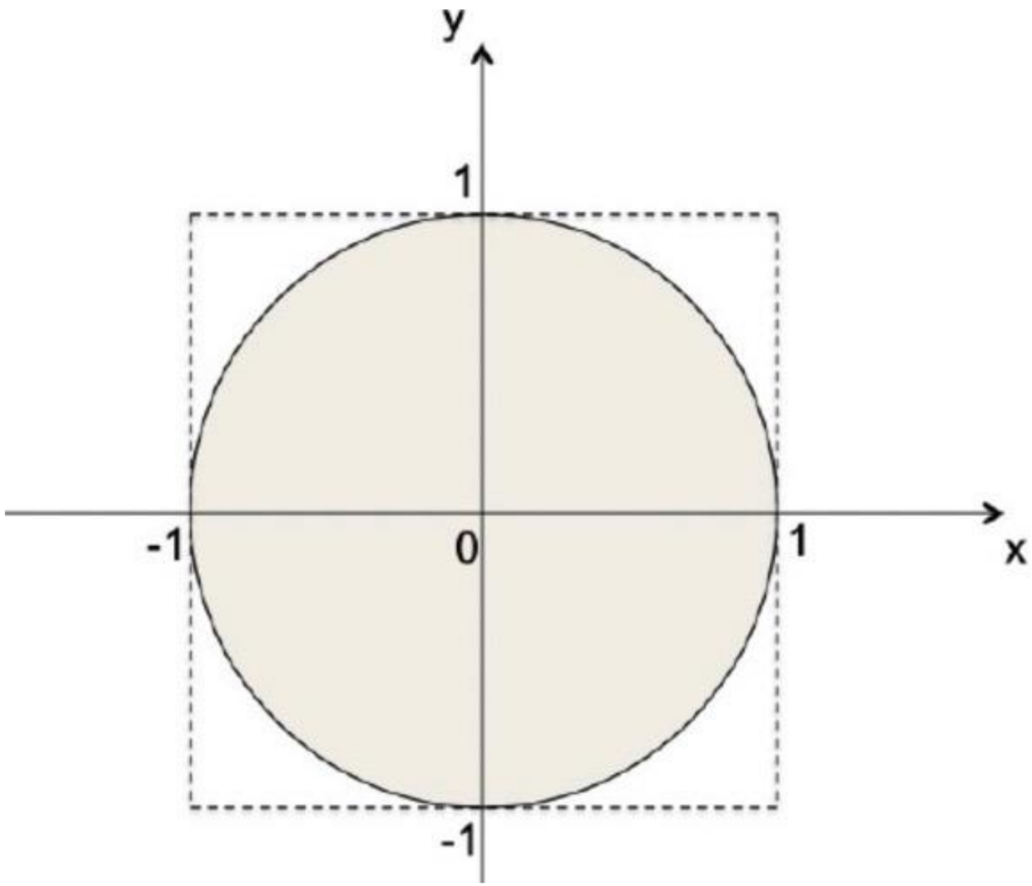
- Estimation of π

Draw a two-dimensional point from the square

$X \sim Uniform(-1,1)$

$Y \sim Uniform(-1,1)$

The sample proportion that $X^2 + Y^2 \leq 1$
is approximately $\pi/4$



Problem Setup:

- You are drawing random points from a square that is centered at the origin with sides of length 2.
- So the points (X, Y) you draw have the following properties:
  - $X \sim Uniform(-1,1)$
  - $Y \sim Uniform(-1,1)$
- Now, consider a circle with radius 1 that is inscribed within this square. The equation of this circle is:

$$X^2 + Y^2 \leq 1$$

Goal:

- We want to estimate π. We know that the area of the circle is $\pi \times r^2 = \pi$, and the area of the square is 2×2=4.
- The ratio of the area of the circle to the area of the square is

$$\frac{\pi}{4}$$

Estimation Procedure:

- **Generate random points**: You generate many random points in the square where both X and Y are uniformly distributed between -1 and 1. So each point (X,Y) falls inside the square.
- **Check if the point is inside the circle**: For each point, check if it satisfies the condition $X^2+Y^2 \leq 1$. If it does, the point lies inside the circle.
- **Compute the proportion**: After generating many points, you can compute the proportion of points that fall inside the circle. Let's call this proportion P. This is the fraction of points where $X^2+ Y^2 \leq 1$.
- **Estimate π**: Since the proportion P is approximately the ratio of the areas of the circle and the square, we have the following relationship:

$$P \approx \frac{\pi}{4}$$

Therefore, to estimate π, you simply multiply the proportion P by 4:

$$\pi \approx 4 \times P$$

- Let's say you generate 1,000 points, and 785 of them satisfy the condition $X^2 + Y^2 \leq 1$. The proportion P is:

$$P = 785/1000 = 0.785$$

- Now, multiply this by 4 to estimate π:

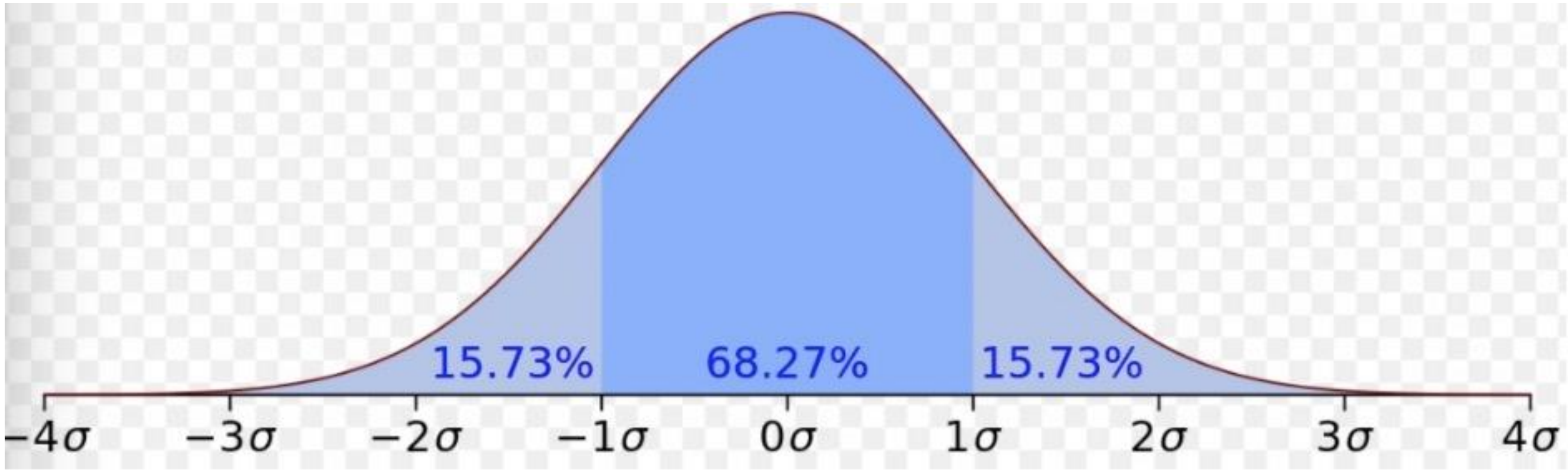
$$\pi \approx 4 \times 0.785 = 3.14$$

Most important:
Normal distribution

- X can be any real number
- Parameters: μ and σ

$$f(x;\mu,\sigma)=\frac{1}{\sigma\sqrt{2\pi}}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)。$$

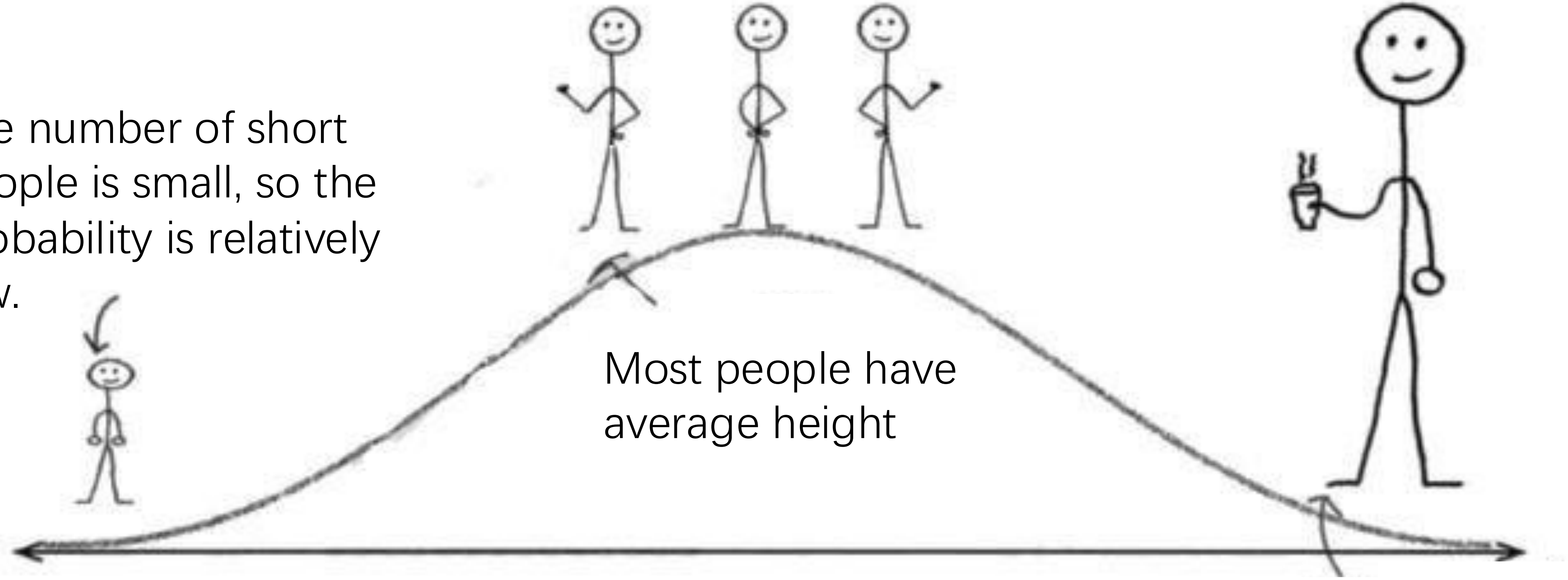
- $X \sim Normal(\mu, \sigma)$



Why we have this distribution?

Normal Distribution: examples

The number of short people is small, so the probability is relatively low.



Most people have average height

There are not many tall people

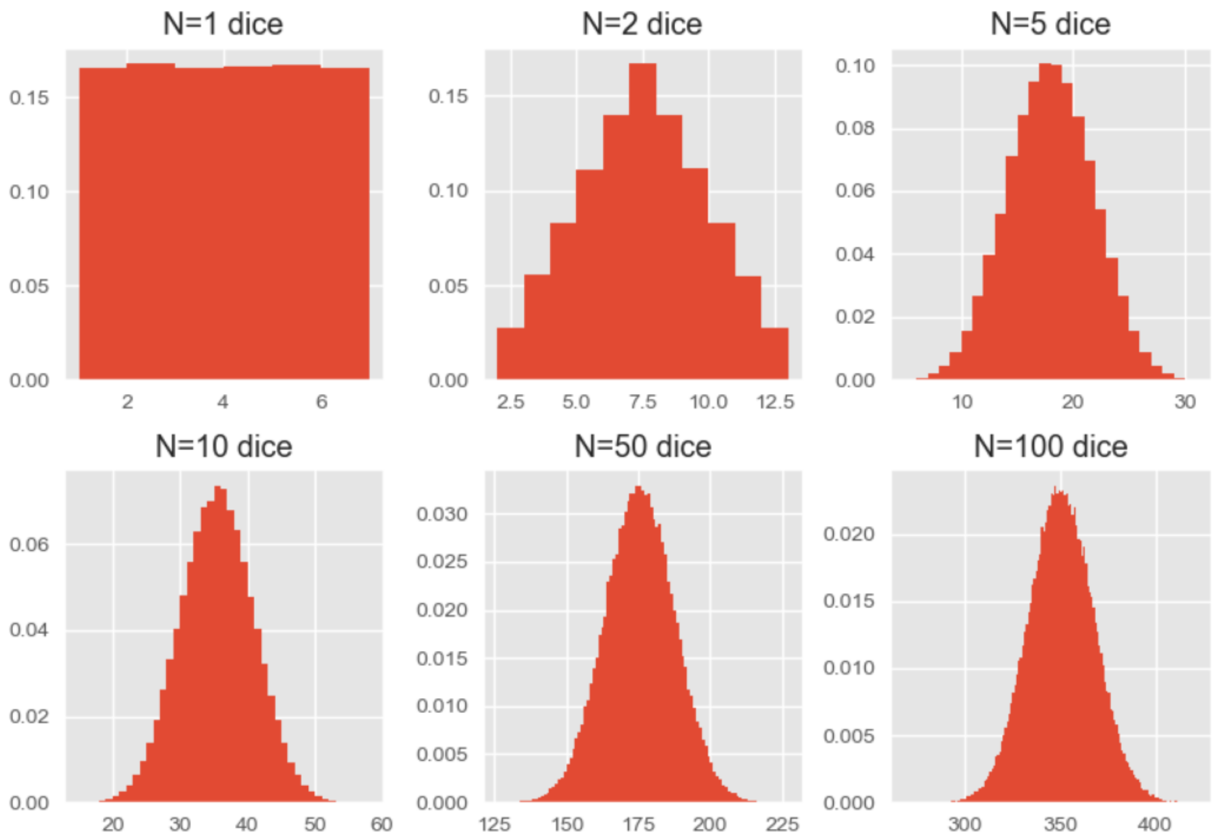
Human Being's Height

Normal Distribution: examples

- A large number of layers
- When a ball goes through each layer, it randomly goes left or right

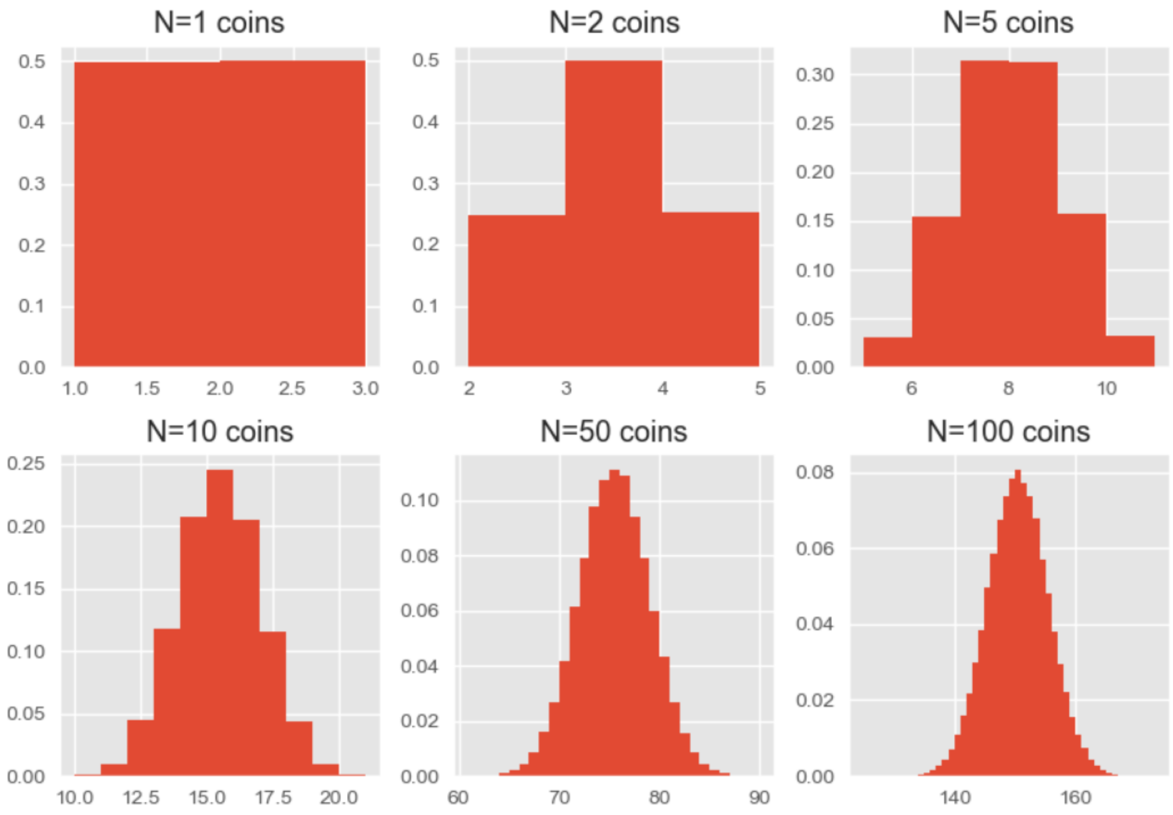
An example

- Toss a die N times
- Let X be the sum
- A demo: N = 1,2,⋯.,100, see the pmf of X



An example

- Flip a fair coin N times
- Let X be the sum (head: 1; tail: 2)
- A demo: N = 1,2,⋯.,100, see the pmf of X



Normal Distribution

Central limit theorem

Lindeberg–Lévy CLT. Suppose $\{X_1, \ldots, X_n\}$ is a sequence of i.i.d. random variables with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}[X_i] = \sigma^2 < \infty$. Then as n approaches infinity, the random variables $\sqrt{n}(\bar{X}_n - \mu)$ converge in distribution to a normal $\mathcal{N}(0, \sigma^2)$:

$$\sqrt{n}\left(\bar{X}_n - \mu\right) \xrightarrow{d} \mathcal{N}\left(0, \sigma^2\right).$$

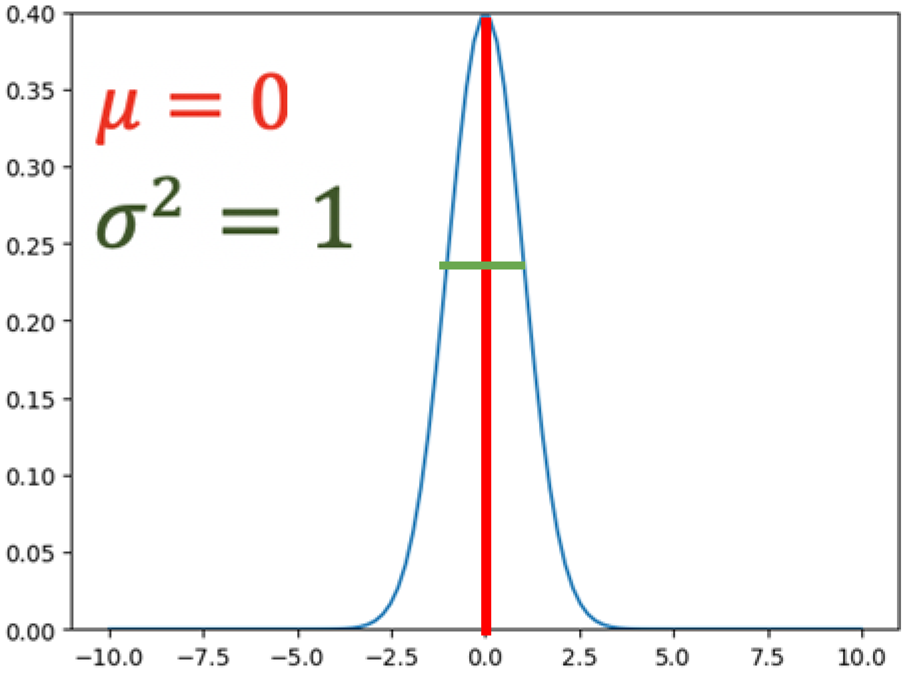
no need to grasp!!!

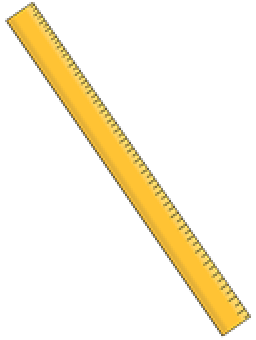
What is the meaning of the parameters?

Mean and Variance

- Mean: μ
- Variance: σ^2

The Normal PDF





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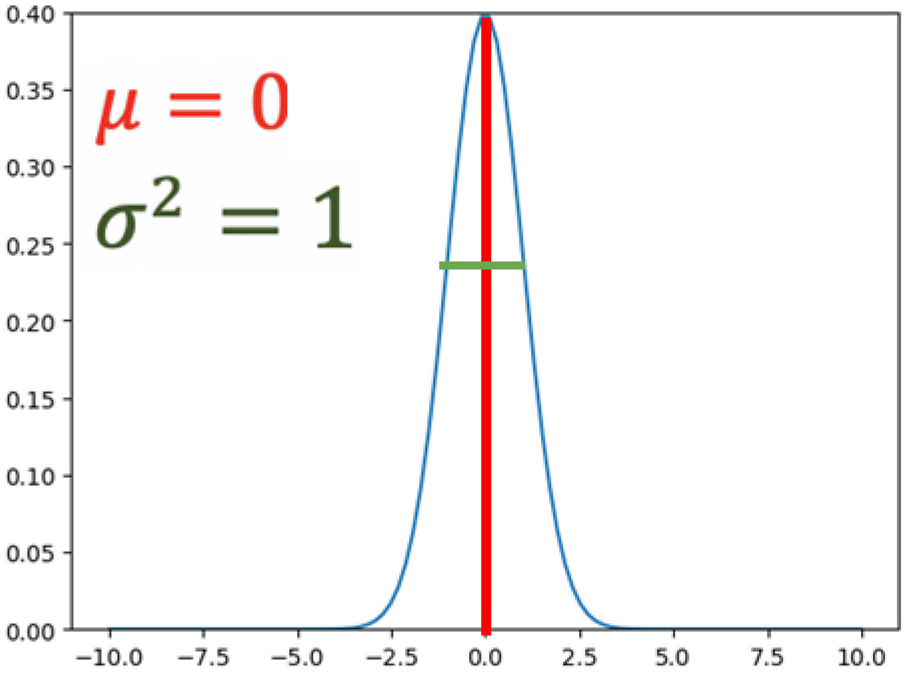

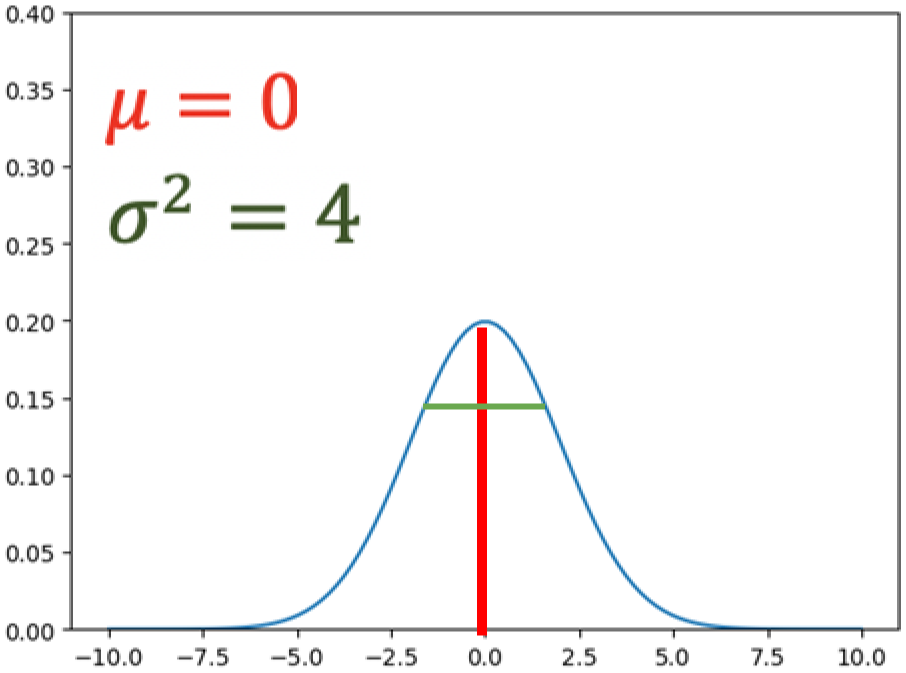

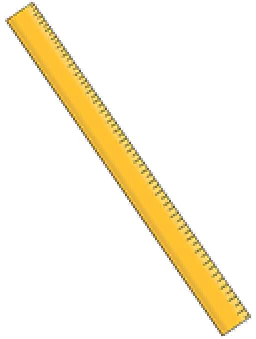

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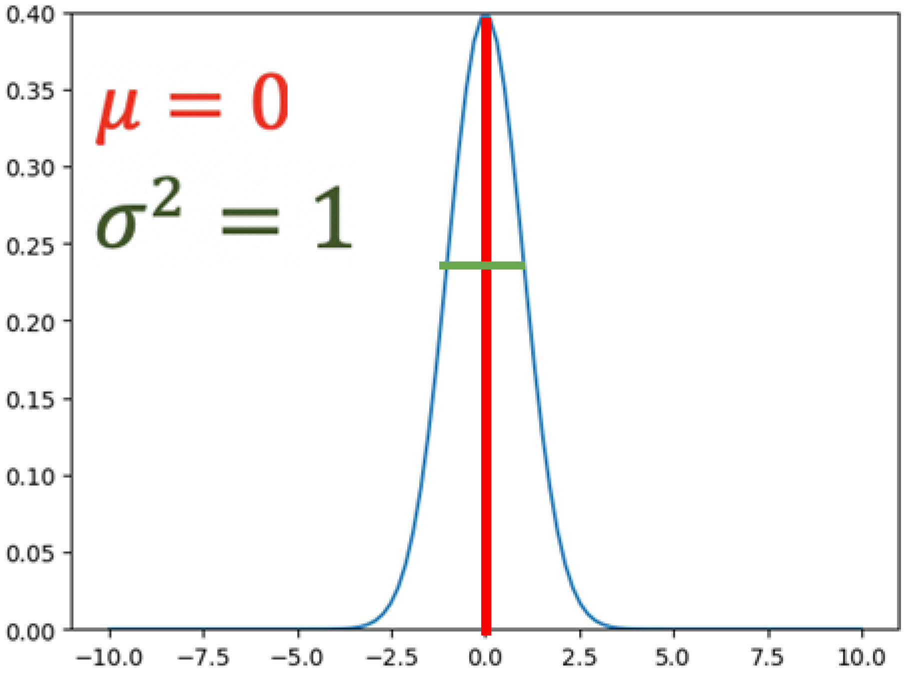

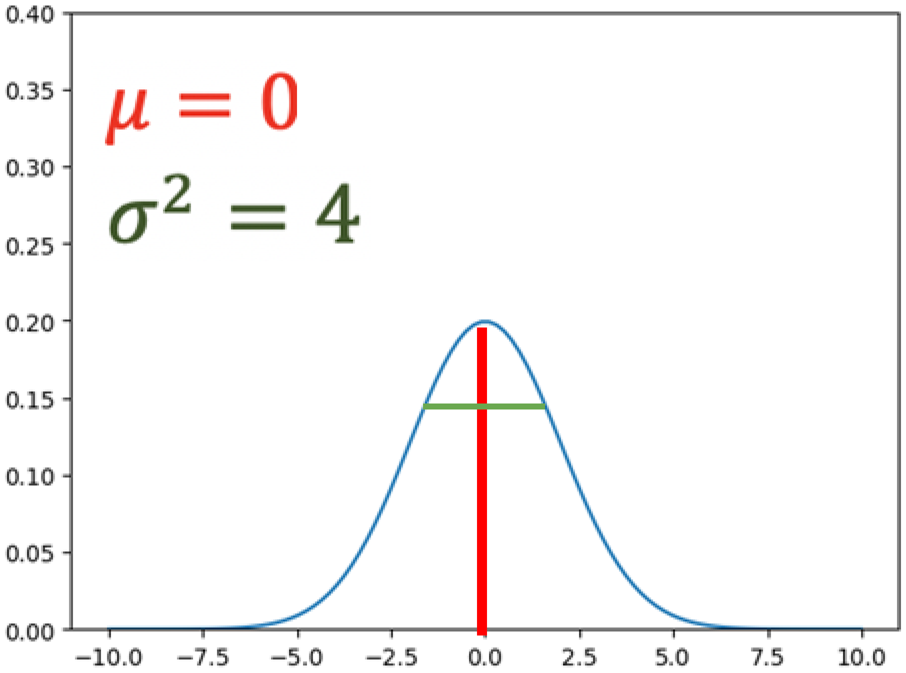

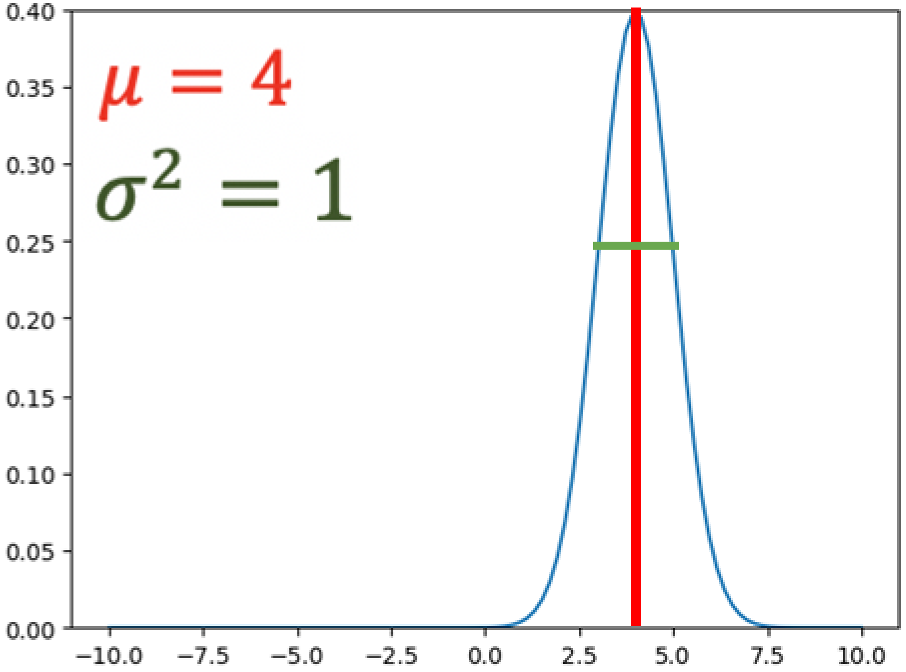

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